Air Quality Data

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1 Origin of the data

This dataset corresponds to air quality monitoring around 3D printers in S.mart Fablab over one year. The air quality is monitored both inside a printer (just before air goes through a filter) and outside it (ambient air around the printer).

The acquisition system monitors every minute temperature, humidity, CO2, levels of particulate matter (PM) and levels of volatile organic compound (VOC). The system is voluntary based on low-cost sensors. The original purposes of the study includes: to monitor air quality according to various parameters, to test the quality of the sensors, to test the efficiency of the filters.

This dataset was created with funding from Idex Formation of Univ. Grenoble Alpes, through a project bringing together Grenoble INP - Génie industriel, IUT Chime – UGA ET S.mart.

2 Description of data

The datasets is split into two similar datasets:

- `EXT` corresponds to a first set of sensors that is supposed to be placed outside the printer, to monitor the air just after it passes through a filter.
- `INT` corresponds to a second set of sensors that is supposed to be placed inside inside the printer, to monitor the air just before it passes through a filter.

2.1 Variables

- `Time` — timestamp of the record (date and time)
- `temperature` — temperature (°C)
- `humidity` — relative humidity (percentage)
- `co2` — carbon dioxyde (CO2) level (ppm)
- `noxr` – nitrogen oxide (NOx) level (raw sensor value)
- `noxi` — nitrogen oxide (NOx) level (constructor-dependent index)
- `voxr` – VOC level as measured by sensor A (raw sensor value)
- `voxi` — VOC level as measured by sensor A (constructor-dependent index)
- `BME680` – VOC level as measured by sensor B (raw sensor value)
- `sfa30` – VOC level as measured by sensor C (raw sensor value)
- `pm10p0` — 10.0 µm particles (µg per m3)
- `pm4p0` — 4.0 µm particles (µg per m3)
- `pm2p5` — 2.5 µm particles (µg per m3)
- `pm1p0` — 1.0 µm particles (µg per m3)

2.2 Known limits

- The position of the sensors (`EXT`, `INT`) is theoretical and sensors have been moved around during the year. In particular, `INT` may monitor ambient air, inside of the air inside the printer, over certain periods of time.
- Filters are sometimes changed, which affects filtering.
- Sensors are low-cost ones and measurements are subject to various kind of errors : gaps, drifts, jumps...
- VOC and NOx sensors record values without a known unit; values reflect a sensor response/sensibility to some compound rather than a precise measurement of that compound. Index values (when they exist) are transformations of raw values provided by sensor manufacturers and are intended to better reflect compound levels.
- The three sensors for VOC may not be sensible to the same compounds.

3 Possible topics of research

- Check and enhance measurement quality. Which sensors are affected by measurement errors? Can measurements errors be detected or even corrected? Can one know when sensors are moved around?
- Qualify VOC sensors. Are they consistent? How are they different? Is some sensor better than the others?
- Qualify filters. How good filters are? What do they filter (and what don't they)? Is efficiency steady? Can one detect filter changes?
- Qualify Air Quality. Is air quality good for persons working in the fablab?
- Events in the fablab. Can one detect when a printer is working? Can one detect the presence of persons around the printer? Are there unexpected events?
- ...

A Some useful commands

- `as.POSIXct(strptime(s, "%Y-%m-%d %H:%M")` — converts character string `s` into a date-time using format (4-digit year)-(2-digit month)-(2-digit day) (2-digit hour):(2-digit minute), e.g., "2024-04-01 08:07".
- `merge(d1, d2, by="Time")` — merge datasets `d1` and `d2` using column `"Time"`. Several optional arguments allow to better specify the columns to use (`by.x`, `by.y`) or the suffix to append to common columns (`suffixes`).

B Air quality and Comfort measurement

Air quality and comfort can be assessed based on temperature, humidity, levels of CO2, levels par particle matters and/or levels of volatile organic compounds (VOC). Figure 1 show samples of such measurements.

Several criteria exist to qualify air quality or comfort. Figure 2 shows ranges of values for some measurements and the associated quality levels. Of course, several measurements can be considered simultaneously to infer a global air quality.

An alternative criterion is the HUMIDEX that qualifies the comfort from temperature and humidity. Details and formula can be found from Wikipedia (French) or (English).

Other criterion for air quality can be considered (in such a case, cite your sources).

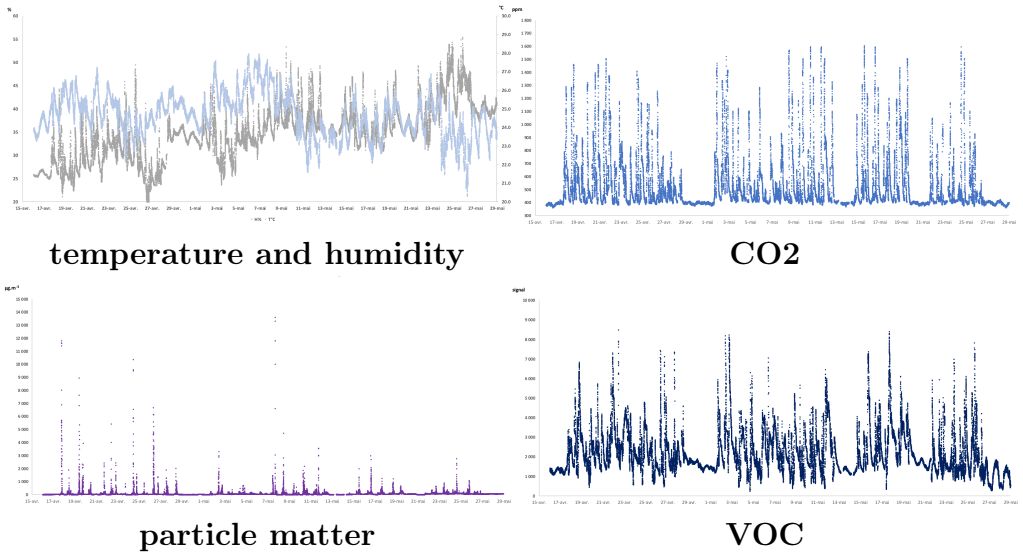


Figure 1: Sample measurements

	Plage CO_2 (ppm)	Plage $PM_{2,5}$ ($\mu g.m^{-3}$)	Plage COV (sans unité)	
0	400 - 600	0 - 25	< 1000	Excellent
1	600 - 1 000	25 - 50	1 000 - 2 000	Bon
2	1 000 - 1 200	50 - 75	2 000 - 3 000	Moyen
3	1 200 - 1 700	75 - 100	3 000 - 4 000	Dégradé
4	1 700- 2 500	100 - 250	4 000 - 5 000	Mauvais
5	> 2 500	> 250	> 5 000	Critique

Figure 2: Ranges for air quality